

Question	Marking Guidance	Mark	Comments
04.1	$\begin{array}{c} \text{R} \\ \\ \text{H}_2\text{N} - \text{C} - \text{COOH} \\ \\ \text{H} \end{array}$	1 (1 x AO1)	Accept other correct representations

Question	Marking Guidance	Mark	Comments
04.2	The primary structure of a protein is the sequence of amino acids joined by peptide bonds.	1 (1 x AO1)	

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04.3	Correct answer of 128 000 000/1.28×10^8/1.3×10^8/1×10^8 = 2 marks;; Answer shows 128 (ignore position of decimal point and preceding/subsequent zeroes, incorrect order of magnitude) = 1 mark OR Evidence of 691 200 000 000/6.912×10^{11}/6.9×10^{11} (total number produced in 24 hours in 5.4 dm³) = 1 mark OR Evidence of $4000 \times 2000 \times 60 \times 60 \times 24$ (total number produced in 24 hours in 5.4 dm³) = 1 mark OR Evidence of 5 333 333.333/5.3×10^6/5×10^6 (total number produced in one hour per cm³ of blood) = 1 mark OR Evidence of 32 000/3.2×10^4 (number produced from one plasma cell per cm³ of blood) = 1 mark;	2 (2 x AO2)	Accept any correct rounding

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04.4	1. Only measuring the number of B cells OR Only provides data for the humoral response OR No data provided for the T cell/cellular response; 2. Only looking at (immune response in) the lungs; 3. Only (measured response to) influenza (virus) OR Only (measured response to) one virus; 4. Day 9 results do not fit the pattern OR Day 9 results are anomalous; 5. Cannot predict if there would be fluctuations between days 11–15/15–19 OR More data needed between days 11–15/15–19 OR Do not know what happens (to the number of B cells) after 19 days; 6. Do not know age (range) of young and old (mice);	3 max (3 x AO3)	2. Accept not looked at other organs 3. Accept need to investigate other viruses/pathogens 3. Accept 'flu' for 'influenza' 5. Accept long term response/effects unknown 5. Accept investigation/ 19 days is too short
